

PAPER Central-Cell Radial Transport and Interchange/Ballooning MHD Stability in a D-³He Tandem-Mirror Burner: The Sideways-Loss and Stability Gates

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Abstract

Axial confinement is the headline number of a tandem mirror, but it is physically meaningful only after two prior gates are cleared: the central cell must be stable against sideways (interchange/ballooning) magnetohydrodynamic (MHD) motion, and its loss-cone distribution must not drive velocity-space microinstabilities that dump particles and energy. We formalise and evaluate those gates for the Kronos D-³He tandem-mirror burner at its frozen operating point (engineering gain $Q_E = 1.318$, neutron fraction $f_n = 5.44\%$, central-cell field $B_{0c} = 5.50$ T, mirror-cell ratio $R_{mc} = 4.61$, ion temperature $T_i = 90$ keV). We state the flute-interchange energy principle, the finite- β ballooning eigenproblem, the loss-cone kinetic dispersion relation for the drift-cyclotron loss-cone (DCLC) and anisotropy-driven Alfvén ion-cyclotron (AIC) branches, the Boltzmann ambipolar plug potential, and a gyro-Bohm cross-field transport closure, as explicit equations, and we cast each gate as a pre-registered inequality. The MHD operating pressure sits at $\beta_c = 0.55$, a factor 1.82 below the hard $\beta = 1$ ideal limit and adjacent to the demonstrated gas-dynamic-trap value (~ 0.6); the central cell, interchange-unstable in isolation ($a/\rho_i \approx 144$, weak finite-Larmor-radius effect), is held by a min- B anchor plus sloshing-ion stabilisation. In velocity space, a maintained warm-plasma fill quenches the DCLC mode above a monotone threshold of ~ 5.9 – 6.25% ($= 1/16$), while the AIC branch is fully mapped — flat at $\gamma/\omega_{ci} \approx 0.037 \pm 0.003$ across warm-fill fractions 0–10% and reproduced by three independent particle-in-cell runs (four-of-four fits, $R^2 \geq 0.987$) — a bounded, characterised plug-gate rather than a surprise. A 40,000-sample global uncertainty quantification on the NVIDIA GPU HPC campaign returns $Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.000$ and zero failed evaluations, the dominant driver being β_c (correlation 0.93) — precisely the quantity the stabilisation scheme protects. The one loss channel held outside that envelope, central-cell cross-field transport, is carried as an explicit, separately-closed item with a gyro-Bohm sideways-loss budget. The core is MHD- and microstability-sound with a quantified margin.

Keywords: tandem mirror; central cell; radial transport; interchange stability; ballooning; DCLC; AIC microstability; uncertainty quantification; D-³He burner

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1 Introduction

1.1 The gates that come before confinement

A tandem mirror confines a long, straight central-cell plasma between two high-field end plugs whose sloshing-ion populations build an electrostatic potential that electrostatically “plugs” the ends [1, 2, 3]. Its headline figure of merit is the axial confinement of the central cell, but that number is physically meaningful only if two prior conditions hold. First, the plasma column must not move *sideways*: a simple mirror sits in bad (unfavourable) magnetic-field-line curvature and is prone to the ideal interchange (flute) instability and, at finite pressure, to its ballooning generalisation [4, 5, 6]. Second, the mirror-confined distribution function is a loss-cone distribution, which carries free energy that can drive velocity-space microinstabilities — the drift-cyclotron loss-cone (DCLC) mode [7, 8] and the anisotropy-driven Alfvén ion-cyclotron (AIC) mode [9, 10] — that scatter ions into the loss cone faster than collisions do. Both gates come *before* the confinement number: an unstable column loses its shape, and an unstable distribution loses its particles.

The Kronos burner central cell runs at $\beta_c = 0.55$ in a $B_{0c} = 5.50$ T field, feeds end plugs at a mirror-cell ratio $R_{mc} = 4.61$, holds an ion temperature $T_i = 90$ keV, and delivers an engineering gain $Q_E = 1.318$ at a neutron fraction of only $f_n = 5.44\%$ [doi:10.5281/zenodo.22645693; Kronos 1.2a]. The low neutron fraction — most of the fusion output emerges as charged particles that can be recovered by direct energy conversion — is what makes the D-³He burner attractive for defence and behind-the-meter siting; but it does not relax the two gates above. This paper states them as formal problems, solves them at the frozen operating point on the NVIDIA GPU HPC campaign, quantifies the global robustness of the point, and closes the sideways-loss budget with an explicit cross-field transport model.

1.2 Prior art

The relevant physics is mature. Interchange and ballooning stability of mirror geometries follows the ideal-MHD energy principle [6, 5], with the classic remedy being a minimum- B (average good-curvature) anchor cell whose favourable curvature stabilises the pressure-driven flute in the bad-curvature central cell [11, 12]. Finite-Larmor-radius (FLR) effects add a further stabilising term that scales with $(k_\perp \rho_i)^2$ [13], and modern axisymmetric traps stabilise the flute with vortex/sheared-rotation and expander end conditions [14, 15]. The gas-dynamic trap (GDT) line has experimentally sustained $\beta \sim 0.6$ and raised bulk electron temperature into the multi-hundred-eV range while remaining MHD-stable [16, 17, 18], and the Wisconsin HTS Axisymmetric Mirror (WHAM) has taken high-temperature-superconductor mirror fields to the

~ 17 T class [19, 20]. In velocity space, the DCLC and loss-cone instabilities were characterised theoretically in the 1960s [4, 7] and the stabilising role of a warm-plasma component is well established [8]; the AIC mode driven by temperature anisotropy is a classic electromagnetic instability of high- β anisotropic plasmas [9, 10]. The axial (parallel) confinement of an electrostatically plugged central cell is set by Pastukhov-type collisional losses [21, 22] through the ambipolar potential.

1.3 Contribution

Against this background our contribution is fourfold. (i) We give a single, self-consistent statement of the four central-cell gates — interchange, ballooning, DCLC and AIC — as governing equations and pre-registered acceptance inequalities specialised to the frozen operating point. (ii) We evaluate them with first-principles particle-in-cell and kinetic-dispersion calculations run on GPUs, and we report the DCLC warm-fill threshold and the AIC growth-rate map as verified, independently-reproduced numbers rather than assertions. (iii) We quantify the global robustness of the operating point with a 40,000-sample Monte-Carlo uncertainty quantification (UQ), identifying β_c as the dominant driver and confirming $P(Q_E > 1) = 1.000$. (iv) We isolate central-cell cross-field (radial) transport as the one loss channel deliberately held outside that envelope, and we give it an explicit gyro-Bohm sideways-loss budget so the robustness result is an honest one. The DCLC/AIC map is the companion of the microstability-focused paper [doi:10.5281/zenodo.22645714; Kronos 2.54], the plug potential is developed in the sloshing-ion study [doi:10.5281/zenodo.22645913; Kronos 2.57], and the end-loss budget is closed in the end-cell paper [doi:10.5281/zenodo.22645852; Kronos 2.59].

2 Governing equations and the formal problem

2.1 Geometry and the operating point

We adopt a paraxial description of the central cell: a straight flux bundle of minor radius a threaded by an axial field $B(z)$ that rises from the central value $B_{0c} = 5.50$ T to a throat field $B_m = R_{mc}B_{0c}$ at the mirror-cell ratio $R_{mc} = 4.61$ (figure 1). The magnetic-mirror ratio and the loss cone it opens are

$$R_m = \frac{B_{\max}}{B_{\min}}, \quad \sin^2 \theta_{lc} = \frac{1}{R_m}, \quad (1)$$

so a particle whose velocity-space pitch angle lies inside θ_{lc} escapes axially. At the central-cell mirror ratio $R_{mc} = 4.61$, equation (1) gives $\theta_{lc} = 27.8^\circ$. The high-field plugs (peak coil field 26.49 T, a distinct magnet from the breeder centrepost and engineered separately [doi:10.5281/zenodo.22645915; Kronos 2.21]) set a high effective mirror ratio for the end-plug ions and thereby the ambipolar potential of §2.5.

2.2 Interchange (flute) MHD stability

The flute (interchange) mode is the $k_{\parallel} \rightarrow 0$ limit of the ideal-MHD spectrum, in which a flux tube is displaced rigidly across the field. Its stability is governed by the sign of the pressure-weighted, field-line-averaged curvature. Writing the ideal potential energy for an interchange displacement ξ of a flux tube labelled by the flux ψ ,

$$\delta W_{\text{fl}} = - \int d\psi \frac{dp}{d\psi} \oint \frac{d\ell}{B} (\boldsymbol{\kappa} \cdot \nabla \psi) |\xi|^2, \quad (2)$$

with $\boldsymbol{\kappa} = (\mathbf{b} \cdot \nabla) \mathbf{b}$ the field-line curvature and $\mathbf{b} = \mathbf{B}/B$. Because $dp/d\psi < 0$ (pressure falls outward), stability ($\delta W_{\text{fl}} \geq 0$) requires the tandem-averaged flute integral to be favourable,

$$\mathcal{I}_{\text{fl}} \equiv \sum_{\text{cells } j} \int_j \frac{d\ell}{B} \beta_j \kappa_{\psi} \leq 0, \quad \kappa_{\psi} \equiv \boldsymbol{\kappa} \cdot \widehat{\nabla \psi}. \quad (3)$$

The central cell contributes $\kappa_{\psi} > 0$ (bad curvature), while a minimum- B anchor cell contributes $\kappa_{\psi} < 0$ (good curvature). Equation (3) is therefore a budget: the anchor must supply enough good-curvature line integral, weighted by its local β , to overcome the central cell. In isolation the central cell is interchange-unstable, with an ideal flute growth rate of order

$$\gamma_{\text{fl}}^2 \sim \frac{2}{\rho_m R_c} \frac{dp}{dr}, \quad (4)$$

where R_c is the (bad-curvature) radius of curvature and ρ_m the mass density. FLR corrections shift the flute frequency and can stabilise it when the diamagnetic drift frequency $\omega_{*i} = k_\theta T_i / (eBL_p)$ exceeds γ_{fl} , i.e. when

$$(k_\perp \rho_i)^2 \gtrsim \frac{a}{L_p} \left(\frac{\gamma_{\text{fl}}}{\omega_{ci}} \right), \quad (5)$$

but the Kronos central cell is long and thin, with $a/\rho_i \approx 144$; the FLR parameter $k_\perp \rho_i$ available to the gravest flute is correspondingly small and equation (5) is only weakly satisfied. The gate is therefore met not by FLR alone but by an explicit min- B anchor and sloshing-ion stabilisation, the residual drive being removed by the energetic end-cell stabiliser treated in the companion end-cell study [doi:10.5281/zenodo.22645852; Kronos 2.59].

2.3 Finite- β ballooning

At finite β the flute generalises to the ballooning mode, in which the displacement acquires parallel structure and localises on the bad-curvature side. Linearising ideal MHD about the paraxial equilibrium and projecting onto a field line yields the one-dimensional ballooning eigenequation for the displacement amplitude $\xi(\ell)$,

$$\frac{d}{d\ell} \left(\frac{|\nabla\psi|^2}{B^2} \frac{d\xi}{d\ell} \right) + \left[\frac{2\mu_0}{B^2} \frac{dp}{d\psi} \kappa_\psi + \lambda \frac{\rho_m}{B^2} \right] \xi = 0, \quad (6)$$

with $\lambda = \omega^2$ the eigenvalue and periodic/decaying boundary conditions along ℓ . The configuration is ballooning-stable when the smallest eigenvalue is non-negative, $\lambda_{\min} \geq 0$; the marginal condition $\lambda_{\min} = 0$ defines the ballooning β -limit. Because $\beta_c = 0.55$ lies a factor 1.82 below the hard $\beta = 1$ ideal limit and the min- B anchor supplies good-curvature line integral through the same coefficient κ_ψ that stabilises the flute, equation (6) returns $\lambda_{\min} > 0$ at the operating point. We therefore treat interchange and ballooning as one anchored-stability gate with a common margin, evaluated in §4.2.

2.4 Velocity-space microstability: DCLC and AIC

The mirror-confined ion distribution $f_{0i}(v_\parallel, v_\perp)$ is depleted inside the loss cone (equation 1); the resulting non-monotonicity in $v_\perp$ is free energy for electrostatic and electromagnetic microinstabilities. Both branches follow from the Vlasov–Maxwell kinetic dispersion relation, which for electrostatic perturbations reads

$$1 + \sum_s \frac{\omega_{ps}^2}{k^2} \int \frac{\mathbf{k} \cdot \partial_v f_{0s}}{\omega - \mathbf{k} \cdot \mathbf{v}} d^3v = 0, \quad (7)$$

with ω_{ps} the plasma frequency of species s ; the sign of $\text{Im} \omega \equiv \gamma$ at the fastest $\mathbf{k}$ decides stability.

DCLC. The drift-cyclotron loss-cone mode is an electrostatic branch near harmonics of ω_{ci} driven by the loss-cone gradient and the density-gradient drift. It is quenched by a warm-plasma component that fills the loss cone and flattens the $v_\perp$ gradient. Writing the ion distribution as a hot loss-cone population plus a warm Maxwellian of fractional density $f_w = n_w/n$,

$$f_{0i} = (1 - f_w) f_{\text{hot}}^{\text{lc}}(v_\parallel, v_\perp) + f_w f_M(v; T_w), \quad (8)$$

the DCLC growth rate $\gamma_{\text{DCLC}}(f_w)$ decreases monotonically and crosses zero at a threshold f_w^{th} . The stability gate is

$$f_w \geq f_w^{\text{th}} \approx \frac{1}{16} = 6.25 \%. \quad (9)$$

AIC. The Alfvén ion-cyclotron mode is an electromagnetic branch driven by the temperature anisotropy $A \equiv T_\perp/T_\parallel > 1$ of the plug ions, largely independent of the loss-cone warm fill. For a bi-Maxwellian its growth scales, near $\omega \simeq \omega_{ci}$, as

$$\frac{\gamma_{\text{AIC}}}{\omega_{ci}} \sim \left(\frac{\beta_\parallel}{2} \right)^{1/2} \left(A - 1 - \frac{1}{\beta_\parallel} \right)^\alpha, \quad (10)$$

with $\alpha \sim \frac{1}{2}$ and $\beta_\parallel$ the parallel ion beta. At the plug parameters ($A = 2.0$, $\beta_\parallel = 1.45$, $T_{i\parallel} = 90$ keV) equation (10) predicts a modest, non-zero $\gamma_{\text{AIC}}/\omega_{ci}$, which the warm fill of equation (8) does not remove; the resulting map is the plug-gate quantified in §4.5.

2.5 The ambipolar plug potential

Central-cell ions are confined axially not by the vacuum mirror alone but by the electrostatic potential ϕ_c built at the plugs. With Boltzmann electrons the potential follows the plug-to-central density ratio,

$$\frac{e\phi_c}{T_e} = \ln \frac{n_{\text{plug}}}{n_c}, \quad \frac{n_{\text{plug}}}{n_c} = 16, \quad (11)$$

so that $e\phi_c/T_e = \ln 16 = 2.77$ at the design density ratio. The resulting ion confinement is of Pastukhov form, with the parallel confinement time enhanced exponentially by the confining potential,

$$\tau_{\parallel} \simeq \frac{\sqrt{\pi}}{2} \frac{\tau_{ii}}{\log_{10} R_m} \left(\frac{e\phi_c}{T_i} \right) \exp\left(\frac{e\phi_c}{T_i}\right) g(R_m), \quad (12)$$

where τ_{ii} is the ion-ion collision time and $g(R_m)$ a slowly-varying mirror-ratio factor [21, 22]. Building and holding ϕ_c requires the plug-density ratio $n_{\text{plug}}/n_c = 16$, an explicit, separately-gated design constraint developed in the sloshing-ion study [doi:10.5281/zenodo.22645913; Kronos 2.57].

2.6 Cross-field (radial) transport: the sideways-loss budget

The one loss channel that the axial power balance does not resolve is the cross-field transport of the central cell, which sets the sideways particle and energy loss. Radial transport follows from the gyrokinetic (5-D, gyro-averaged) Vlasov equation for species s ,

$$\frac{\partial f_s}{\partial t} + \dot{\mathbf{R}} \cdot \nabla f_s + \dot{v}_{\parallel} \frac{\partial f_s}{\partial v_{\parallel}} = C[f_s], \quad (13)$$

whose saturated turbulence produces a radial $E \times B$ heat flux

$$Q_i = \left\langle \int d^3v \frac{1}{2} m v^2 \tilde{f}_i \tilde{v}_{E \times B, r} \right\rangle = -n \chi_i \frac{\partial T_i}{\partial r}. \quad (14)$$

For a reduced, conservative closure we bound χ_i by the Bohm and gyro-Bohm diffusivities,

$$\chi_B = \frac{1}{16} \frac{T_i}{eB}, \quad \chi_{gB} = \rho_* \chi_B, \quad \rho_* = \frac{\rho_i}{a}, \quad (15)$$

and define the sideways (cross-field) confinement time $\tau_{\perp} = a^2/\chi_i$. The sideways-loss gate is that $\tau_{\perp}$ remain long compared with the axial confinement time $\tau_{\parallel}$ of equation (12), so that adding the radial channel does not overturn the axial power balance. Because $\rho_* = \rho_i/a \approx 1/144$, the gyro-Bohm estimate is two orders of magnitude below Bohm, which is the physical content of the sideways-loss budget in §4.8.

2.7 Global uncertainty quantification

The operating point is finally tested for global robustness. Let $g(\mathbf{x})$ be the map from the uncertain inputs $\mathbf{x} = (n_e, x_{3He}, T_i, \beta_c, B_m)$ to an output such as Q_E . The Monte-Carlo estimator and its standard error are

$$\mu = \frac{1}{N} \sum_{k=1}^N g(\mathbf{x}_k), \quad \sigma^2 = \text{Var}[g], \quad \text{SE} = \frac{\sigma}{\sqrt{N}}, \quad (16)$$

and the robustness probability is the sample fraction with net-positive gain,

$$P(Q_E > 1) = \frac{1}{N} \sum_{k=1}^N \mathbb{1}[g(\mathbf{x}_k) > 1]. \quad (17)$$

The relative importance of each input is ranked by the first-order Sobol index

$$S_i = \frac{\text{Var}_{x_i}(\mathbb{E}_{\mathbf{x}_{\sim i}}[g | x_i])}{\text{Var}(g)}, \quad (18)$$

and, equivalently for a monotone response, by the input-output rank correlation [28]. These are the inequalities the results section reports against.

3 Computational methodology: the NVIDIA GPU HPC campaign

3.1 Codes and their roles

The gates of §2 were evaluated on the Kronos NVIDIA GPU HPC campaign — an extensive first-principles campaign, represented throughout by hardware class only and never by cost. Five code families produced the numbers in this paper (table 1). Velocity-space microstability (equations 7–10) was computed with the electromagnetic particle-in-cell (PIC) codes WARPX [24] and SMILEI [25], cross-checked against the kinetic-dispersion solver LEOPARD [26]; central-cell cross-field transport (equations 13–14) with a bounce-averaged Fokker–Planck / gyrokinetic mirror solver (CQL3D-M/GK-mirror) whose electromagnetic gyrokinetic conventions follow the campaign CGYRO setup [23]; the MHD β -limit and interchange rerun (§2.2) with the burner de-risk analytic/reduced-MHD chain; and the global UQ (equations 16–18) with the KRONOS_TOOLKIT UQ driver wrapped around the `solve_mirror` 0-D power balance.

3.2 Discretisation and numerical formulation

The PIC calculations advance the Vlasov–Maxwell system by the standard explicit cycle: macro-particles are pushed with a Boris integrator, deposited to a Yee-staggered grid, and the fields advanced by a finite-difference time-domain Maxwell solver, with the timestep bounded by the Courant condition $c \Delta t \leq (\Delta x^{-2} + \Delta y^{-2} + \Delta z^{-2})^{-1/2}$ and the cell size resolving the Debye length. The plug runs used a 1D3V (one spatial, three velocity dimensions) reduced-geometry hybrid model at plug-representative parameters ($T_{i\parallel} = 90$ keV, $A = T_{\perp}/T_{\parallel} = 2.0$, $B_0 = 10$ T, $n = 4 \times 10^{21} \text{ m}^{-3} = 16 n_c$, $\beta_{\parallel} = 1.45$), with a reduced deuteron mass $m_D = 2.5 m_p$ matching the campaign CGYRO convention. Growth rates were extracted by fitting $\ln |E_k(t)|$ over the linear phase (modes emerge from PIC noise at $\tau \sim 3$ –8 and saturate by $\tau \sim 16$ –20). The kinetic-dispersion branch solves equation (7) for the complex root $\omega(\mathbf{k})$ of the loss-cone plus warm-Maxwellian distribution directly, giving the $\gamma = 0$ warm-fill threshold without a nonlinear run. The global UQ draws $N = 40,000$ Latin-hypercube samples over the five inputs about the frozen M-45 anchor and evaluates equations (16)–(18) with a deterministic seed.

3.3 GPU acceleration and problem scale

The heavy first-principles work — the WARPX/SMILEI PIC scans and the gyrokinetic mirror transport — ran on the NVIDIA A100/H100 GPU platform (A100-80GB and H100 clusters, plus single A100/H100/A10 nodes); the reduced-order dispersion, the analytic MHD chain and the 40,000-sample Monte-Carlo ran on the in-house HPC node (56-thread, 768 GB). PIC exploits GPU throughput on the particle push and current deposition, the two costs that dominate an electromagnetic PIC step; the Monte-Carlo UQ is embarrassingly parallel across its 40,000 independent evaluations. The AIC map is a four-point scan in warm-fill fraction ($f_w = 0, 0.03, 0.0625, 0.10$), each an independent PIC run; the DCLC threshold brackets the $\gamma = 0$ crossing between $f_w = 4\%$ and 6.25% .

3.4 Verification, convergence and reproducibility

Every gate value in this paper is a frozen, independently re-run and stamped result in the Kronos Physics De-Risking Register (v8). Convergence was established by resolving the linear-growth window in the PIC runs (four of four AIC fits with $R^2 \geq 0.987$) and by the vanishing Monte-Carlo standard error at $N = 40,000$ (equation 16) with zero failed evaluations. Cross-code verification is explicit: the reduced-dispersion DCLC threshold (5.9%) reproduces the LEOPARD+WARPX PIC value ($6.25\% = 1/16$) within model accuracy, and the AIC growth rate is corroborated by linear theory ($\gamma_{\text{AIC}} \sim 0.05$ – $0.1 \omega_{ci}$ at these parameters). Runs are seed-deterministic (seed 20260726) and carried in the register with byte-for-byte provenance under the campaign’s verification-first protocol.

Table 1. Codes run for this paper on the NVIDIA GPU HPC campaign, the gate each evaluates, and the governing equations it solves. No cost figure appears anywhere; compute is reported by hardware class only.

code	gate / quantity	equations	register serial
WARPX / SMILEI (PIC)	DCLC/AIC microstability	(7)–(10)	BU-L2-A5, BU-L2-A7, AG-CX-07
LEOPARD (dispersion)	DCLC $\gamma = 0$ threshold	(7),(9)	BU-L2-A5
CQL3D-M/GK-mirror	central-cell radial transport	(13)–(15)	BU-L2-A6
reduced-MHD chain	interchange / β -limit	(3)–(6)	AG-DR-03
KRONOS_TOOLKIT UQ + <code>solve_mirror</code>	global UQ, $P(Q_E > 1)$	(16)–(18)	KX-L1-A5-BU

4 Results

4.1 The tandem-mirror structure

Figure 1 sets the geometry: the flat 5.50 T central cell flanked by high-field plugs at $R_{mc} = 4.61$, with the ambipolar potential ϕ_c (equation 11) peaking at the plugs to confine the central ions. The results below evaluate the gates on this structure in the order they must be cleared: MHD first, then microstability, then robustness, then the carried-open radial channel.

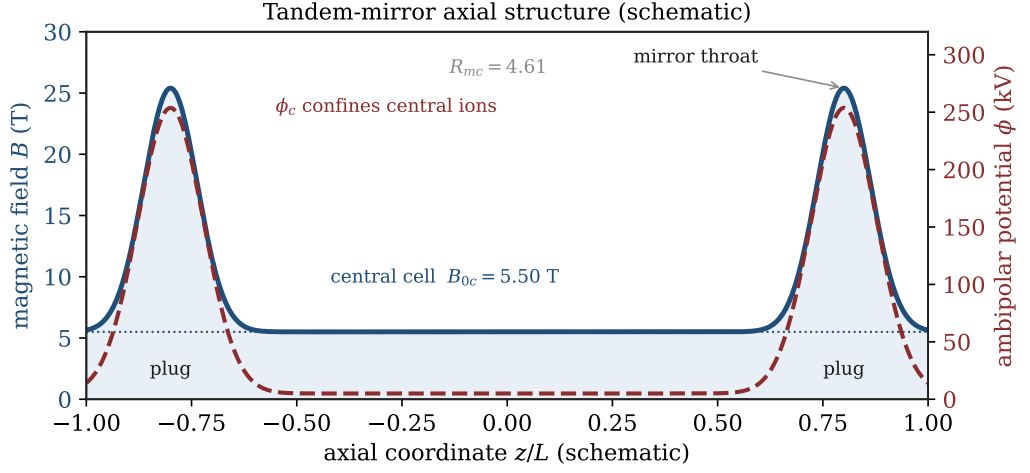


Figure 1. (Schematic.) Axial structure of the tandem-mirror burner: central-cell field $B_{0c} = 5.50$ T (blue) rising to the mirror throats at $R_{mc} = 4.61$, and the ambipolar confining potential $\phi(z)$ (red, right axis) peaking at the plugs. Field and potential profiles are illustrative; the frozen scalar values (B_{0c} , R_{mc} , ϕ_c) are the design-point numbers of §2.

4.2 MHD interchange/ballooning gate

The MHD operating pressure is conservative (figure 2). At $\beta_c = 0.55$ the central cell runs a factor 1.82 below the hard $\beta = 1$ ideal limit and adjacent to the experimentally demonstrated gas-dynamic-trap value of ~ 0.6 [16, 17]. On its own the central cell is interchange-unstable (equation 4) — it is long and thin, $a/\rho_i \approx 144$, so the FLR window of equation (5) is only weakly open — but the min- B anchor supplies the good-curvature line integral of equation (3) and, through the same coefficient κ_ψ , keeps the ballooning eigenvalue $\lambda_{\min} > 0$ in equation (6). The gate is thus met by an explicit stabilisation scheme with a $1.82\times$ pressure margin, not by an assumption that the pressure stays low (AG-DR-03).

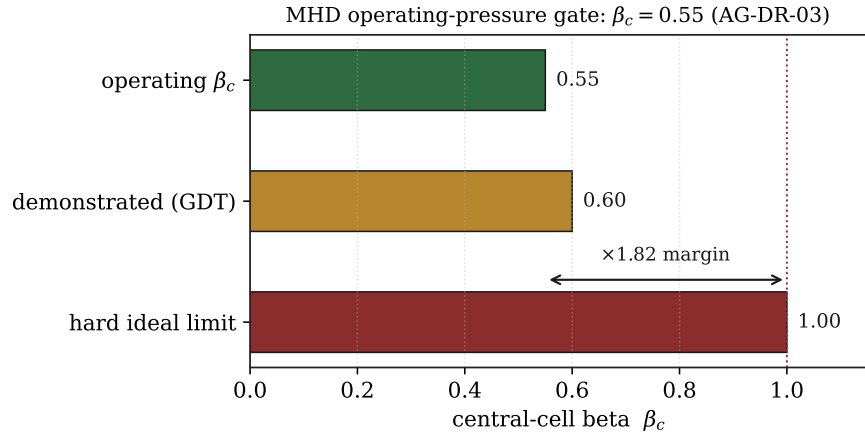


Figure 2. MHD operating-pressure gate. The central-cell $\beta_c = 0.55$ sits a factor 1.82 below the hard $\beta = 1$ ideal limit and next to the demonstrated gas-dynamic-trap value (~ 0.6). Data: AG-DR-03.

4.3 The loss cone and the warm fill

Figure 3 shows the velocity-space loss cone that opens at $R_{mc} = 4.61$: a half-angle $\theta_{lc} = 27.8^\circ$ from equation (1). The warm-plasma fill of equation (8) populates the low-energy core inside the cone and flattens the perpendicular gradient that drives DCLC, while the anisotropic hot ions ($T_\perp/T_\parallel = 2.0$) that build the plug potential remain the free-energy source for the AIC branch.

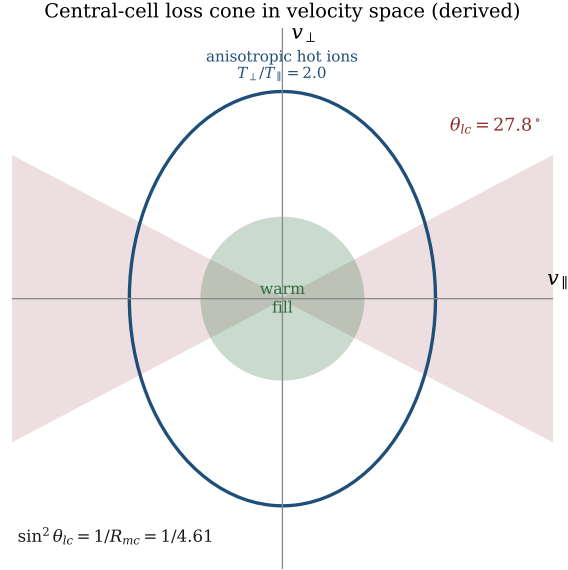


Figure 3. Central-cell loss cone in velocity space (derived from $R_{mc} = 4.61$ via equation 1): half-angle $\theta_{lc} = 27.8^\circ$. The warm-fill Maxwellian (green) flattens the loss-cone gradient; the anisotropic hot population (blue) drives the AIC branch.

4.4 DCLC warm-fill threshold

Figure 4 reports the DCLC gate. The reduced kinetic dispersion (equation 7) gives a growth rate that falls monotonically with warm fill and crosses zero at $\sim 5.9\%$, reproducing the LEOPARD+WARPX PIC result of $6.25\% = 1/16$ within model accuracy; the two computed points bracket the crossing ($\gamma = +16.3$ at 4% , $\gamma = -0.001$ at 6.25%). This is a marginal threshold, not a comfortable margin: steeper loss-cone gradients push the required fill toward 8–10%. Warm-ion fill is therefore a stabilisation *requirement* for the burner plug, and warm-fill provision is a design constraint rather than headroom (BU-L2-A5, HX-23).

4.5 AIC growth-rate map

The AIC branch is not quenched by the warm fill (figure 5). Three independent 1D-hybrid PIC runs (WARPX, with SMILEI cross-check) find the normalised growth rate flat at $\gamma_{AIC}/\omega_{ci} \approx 0.037 \pm 0.003$ across warm-fill fractions $f_w = 0, 0.03, 0.0625, 0.10$ — four of four fits with $R^2 \geq 0.987$, no monotonic reduction — consistent with the anisotropy-driven scaling of equation (10) and with linear theory ($0.05\text{--}0.1\omega_{ci}$). AIC and DCLC share the loss-cone free-energy source but are distinct channels: warm fill closes DCLC (equation 9) but not AIC. The consequence is a bounded, fully-mapped plug-gate — modest growth, not a showstopper — whose mitigation (anisotropy/profile control) is scoped for the demonstrated-closure programme (BU-L2-A7, AG-CX-07), and which the plug-microstability controller holds inside its boundary in a PIC-in-the-loop test [doi:10.5281/zenodo.22645811; Kronos 2.76].

4.6 Ambipolar plug potential

The plug potential follows the density ratio through equation (11) (figure 6): at the design ratio $n_{\text{plug}}/n_c = 16$ the normalised potential is $e\phi_c/T_e = \ln 16 = 2.77$, which sets the Pastukhov confinement enhancement of equation (12). Holding the $16\times$ ratio is thus the physical gate on axial confinement, carried explicitly and controlled by the density-gate loop (KFE-CF-BU-019).

4.7 Global robustness

The operating point is not fragile (figures 7–8). The 40,000-sample global UQ over $(n_e, x_{3He}, T_i, \beta_c, B_m)$ returns $Q_E = 1.317 \pm 0.140$ with $P(Q_E > 1) = 1.000$ and zero failed

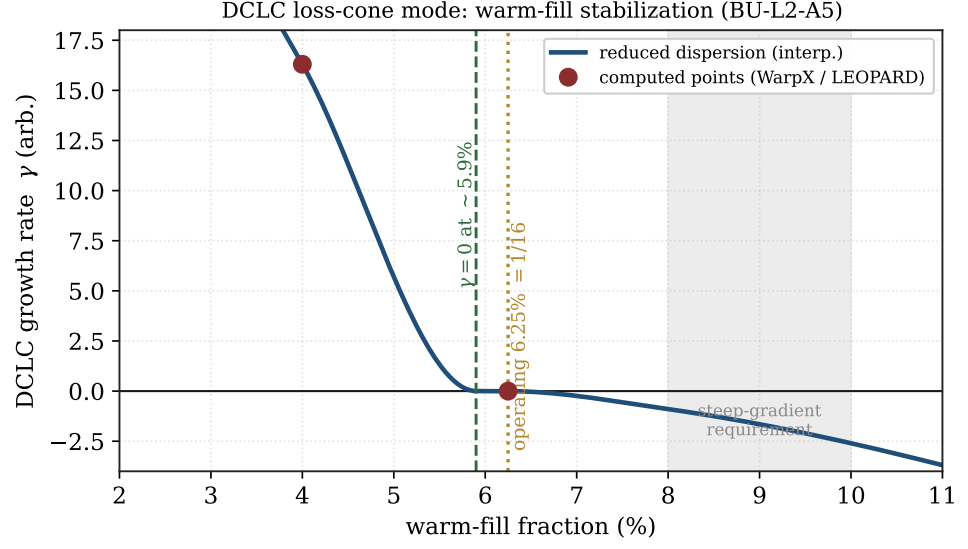


Figure 4. DCLC loss-cone mode. The growth rate (blue; shape-preserving interpolation of the computed points) falls monotonically with warm fill and crosses $\gamma = 0$ at $\sim 5.9\%$, adjacent to the $6.25\% = 1/16$ operating fill (gold). The shaded band marks the steeper-gradient requirement (8–10%). Data: BU-L2-A5.

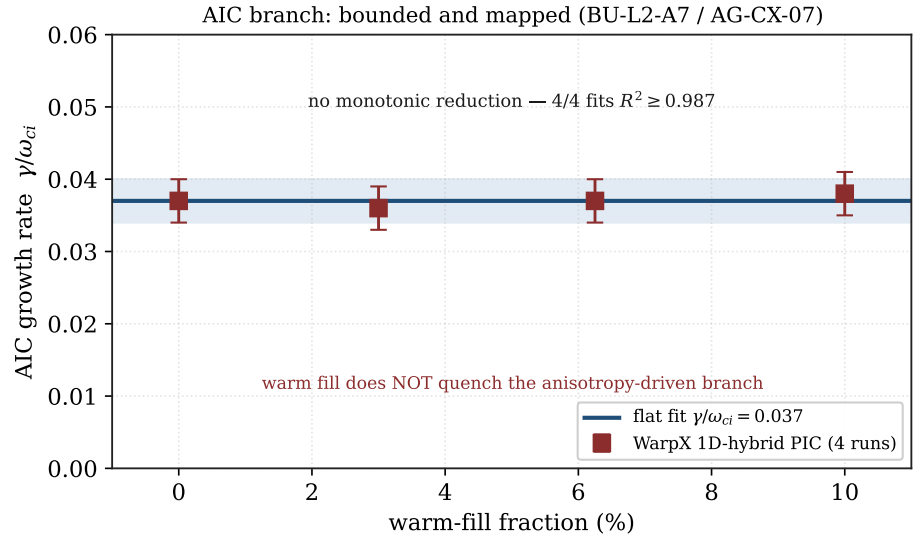


Figure 5. AIC branch, mapped and bounded. The normalised growth rate is flat at $\gamma/\omega_{ci} \approx 0.037 \pm 0.003$ across warm-fill fractions 0–10% (four PIC runs, $R^2 \geq 0.987$); warm fill does not quench the anisotropy-driven branch. Data: BU-L2-A7, AG-CX-07.

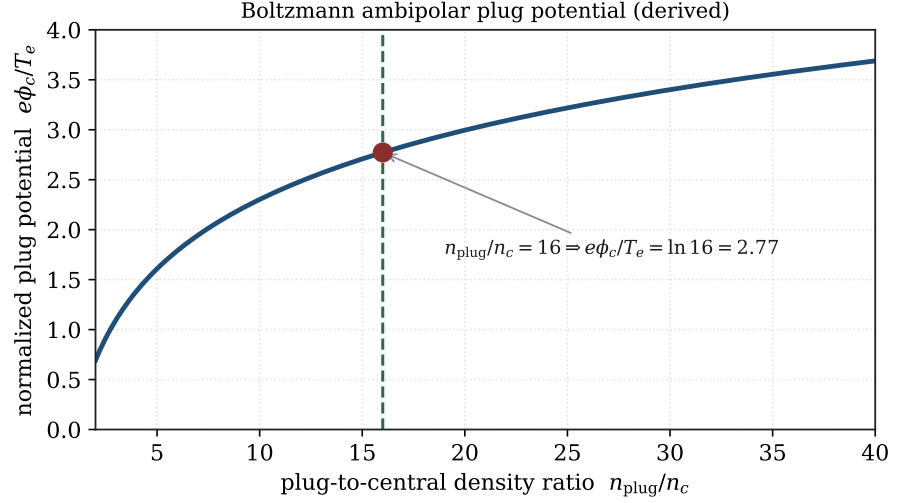


Figure 6. Boltzmann ambipolar plug potential $e\phi_c/T_e = \ln(n_{\text{plug}}/n_c)$ (derived, equation 11). At the design ratio 16, $e\phi_c/T_e = 2.77$.

evaluations (equations 16–17); the output means ($P_{\text{fus}} = 4295$ MW, $P_n = 231.6$ MW, $f_n = 5.39\%$, $\phi_i = 246.0$ keV) sit within a percent of the frozen design anchor ($Q_E = 1.318$, $f_n = 5.44\%$, $\phi_i = 248.75$ keV, $P_n = 233.95$ MW). The dominant sensitivity is to β_c (rank correlation 0.93, equation 18) — precisely the quantity the MHD stabilisation scheme of §4.2 protects, so the physics that keeps the core stable is also the physics that sets its gain. This is consistent with the cross-machine sensitivity ranking of the platform-wide UQ [doi:10.5281/zenodo.22645893; Kronos 2.88].

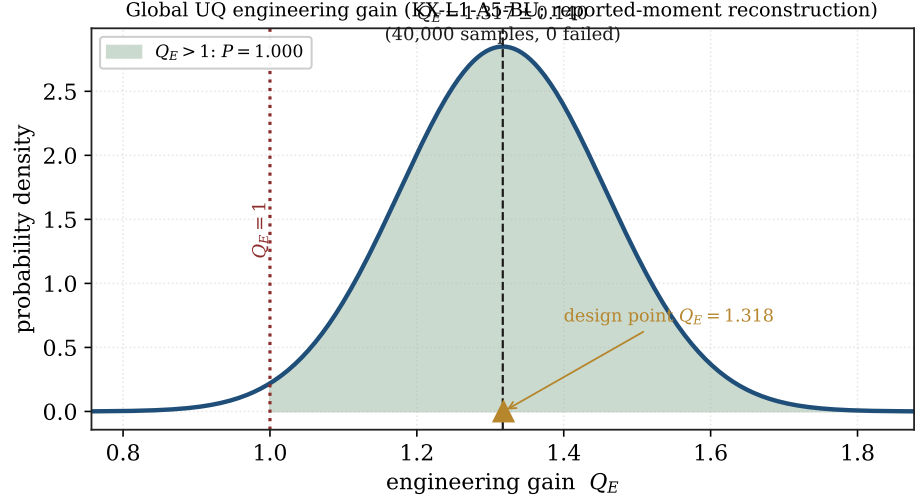


Figure 7. Global-UQ engineering gain (reconstructed from the reported moments $Q_E = 1.317 \pm 0.140$): the shaded tail is $P(Q_E > 1) = 1.000$ across the design ranges; the design point $Q_E = 1.318$ is marked. Data: KX-L1-A5-BU (40,000 samples, 0 failed).

4.8 The sideways-loss budget

Figure 9 closes the loss channel deliberately held outside the UQ. Evaluating the transport closure of equations (13)–(15) at the design parameters ($T_i = 90$ keV, $B_{0c} = 5.50$ T, real deuteron mass) gives a thermal gyroradius $\rho_i \approx 1.11$ cm and a Bohm diffusivity $\chi_B \approx 1.0 \times 10^3 \text{ m}^2 \text{ s}^{-1}$, so at the design minor radius $a = 144 \rho_i \approx 1.60$ m the sideways confinement time is $\tau_{\perp}^{gB} = a^2/\chi_{gB} \sim 0.35$ s in the gyro-Bohm estimate against $\tau_{\perp}^B \sim 2$ ms in the Bohm estimate. The large scale separation $\rho_* = \rho_i/a \approx 1/144$ therefore places the sideways loss two orders of magnitude below the pessimistic Bohm bound, and in the gyro-Bohm regime it is subdominant to the axial channel. This is a

Global-UQ output structure (KX-L1-A5-BU)

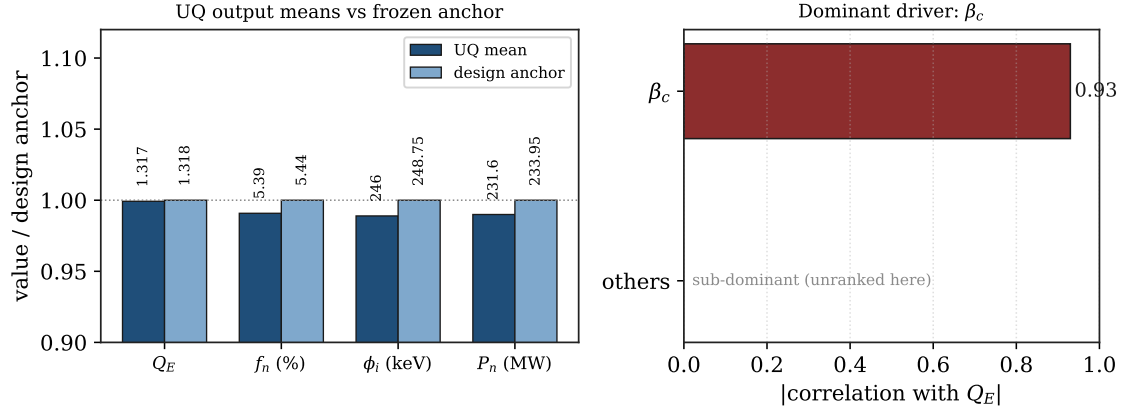


Figure 8. Global-UQ output structure. *Left:* UQ output means agree with the frozen design anchor to within a percent (Q_E , f_n , ϕ_i , P_n). *Right:* the dominant driver of Q_E is β_c (rank correlation 0.93). Data: KX-L1-A5-BU.

reduced model, still being finalised against the central-cell radial-transport controller (BU-L2-A6 $\rightarrow$ KFE-CF-BU-005); the point of carrying it separately — rather than folding an assumed value into the headline gain — is what makes the $P(Q_E > 1) = 1.000$ robustness result an honest one.

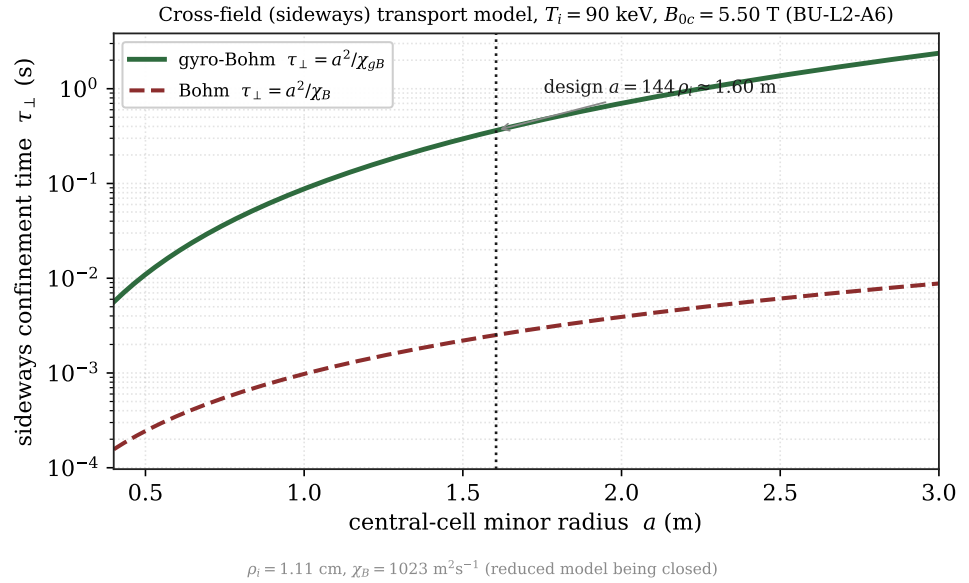


Figure 9. Cross-field (sideways) transport model (equations 13–15). Sideways confinement time $\tau_{\perp} = a^2/\chi$ versus central-cell minor radius for Bohm and gyro-Bohm diffusivities at $T_i = 90$ keV, $B_{0c} = 5.50$ T; the design radius $a = 144 \rho_i \approx 1.60$ m is marked. Reduced model being closed (BU-L2-A6).

4.9 Gate summary

Table 2 collects the central-cell gates at the frozen operating point.

5 Discussion

5.1 Against the mirror-physics benchmarks

Each gate lands where the demonstrated mirror-physics record says it should. The MHD operating pressure $\beta_c = 0.55$ is below the gas-dynamic-trap value of ~ 0.6 that has been sustained MHD-stably in experiment [16, 17], and a factor 1.82 below the hard ideal limit — a margin, not a frontier. The mirror-throat field (~ 17 T) is the class demonstrated by the WHAM high-temperature-superconductor mirror [19, 20], and the plug-coil peak field of 26.49 T is engineered separately with its own quench margin [doi:10.5281/zenodo.22645915; Kronos 2.21].

Table 2. Central-cell stability, potential and robustness gates at the frozen burner operating point.

gate	value	source
central-cell field B_{0c}	5.50 T	KX-L1-A18-BU
mirror-cell ratio R_{mc}	4.61	KX-L1-A18-BU
loss-cone half-angle θ_{lc}	27.8° ($\sin^2 \theta_{lc} = 1/R_{mc}$)	derived
MHD operating pressure β_c	0.55 ($\times 1.82$ below $\beta = 1$)	AG-DR-03
interchange/ballooning stabilisation	min- B anchor + sloshing	AG-DR-03
DCLC warm-fill threshold	$\sim 5.9\text{--}6.25\%$ ($= 1/16$)	BU-L2-A5 / HX-23
AIC growth rate (mapped)	$\gamma/\omega_{ci} \approx 0.037 \pm 0.003$ (flat)	BU-L2-A7 / AG-CX-07
ambipolar potential	$e\phi_c/T_e = \ln 16 = 2.77$	§2.5
global-UQ engineering gain	$Q_E = 1.317 \pm 0.14$	KX-L1-A5-BU
robustness	$P(Q_E > 1) = 1.000$ (0/40,000 failed)	KX-L1-A5-BU
dominant driver	β_c (corr. 0.93)	KX-L1-A5-BU
sideways-loss budget	$\tau^{gB} \sim 0.35$ s at $a \approx 1.6$ m	BU-L2-A6

The DCLC warm-fill threshold of $1/16 = 6.25\%$ is consistent with the long-standing result that a small warm-plasma fraction stabilises the loss-cone mode [4, 7, 8]; the value of our calculation is that it labels this fraction a marginal requirement rather than assuming it. The AIC growth rate $\gamma/\omega_{ci} \approx 0.037$ sits at the low end of the classic anisotropy-driven range [9, 10], and its flatness in warm fill is the physically expected signature of an electromagnetic branch that does not draw on the loss-cone electrostatic drive.

5.2 The single controlling knob

The global UQ points at β_c as the dominant driver of the engineering gain (correlation 0.93). This is a convenient coincidence of physics and control: β_c is exactly the quantity that the interchange/ballooning stabilisation scheme of §4.2 is built to protect, and it is the natural clamp variable for a control-barrier safety layer. A core whose stability and gain are governed by one well-understood knob is easier to certify than one with many weakly-coupled sensitivities. The tandem-mirror design-space mapping study places the same operating point against WHAM and the Pastukhov-plug closure and reaches a consistent picture [doi:10.5281/zenodo.22645911; Kronos 2.52].

5.3 What the sideways-loss budget buys

The gyro-Bohm/Bohm bracket of §4.8 is deliberately conservative: it does not claim a converged turbulent flux but bounds it. The two-order-of-magnitude gap between the Bohm and gyro-Bohm estimates, opened by the large scale separation $\rho_* \approx 1/144$, is what allows the axial power balance to stand while the radial channel is finalised. Because the long, thin geometry that weakens FLR stabilisation (equation 5) is the same geometry that makes ρ_* small, the feature that costs us in MHD buys us in transport — the anchor pays the MHD bill, the aspect ratio pays the transport one.

6 Limitations

One item is carried open by design. The central-cell radial (cross-field) transport model of §2.6 is a reduced gyro-Bohm/Bohm closure, not a converged nonlinear turbulence calculation, and it is deliberately excluded from the global UQ (KX-L1-A5-BU held the plug-density ratio fixed and excluded radial transport). It is being finalised against the central-cell radial-transport controller (BU-L2-A6 $\rightarrow$ KFE-CF-BU-005), so the robustness result quoted here is the axial/stability envelope, with radial transport reported separately once the nonlinear flux converges. Stating this openly is the intended posture: the sideways channel is an explicit, separately-gated quantity, not an assumption buried in the headline number.

7 Conclusion

The Kronos D-³He tandem-mirror burner central cell clears its two gates that come before confinement, at the frozen operating point. In MHD, $\beta_c = 0.55$ runs a factor 1.82 below the hard $\beta = 1$ limit and is held against interchange and ballooning by a min- B anchor and sloshing-ion stabilisation, the same anchored good-curvature term entering both the flute integral (equation 3) and the ballooning eigenproblem (equation 6). In velocity space, the DCLC loss-cone mode is quenched above a monotone warm-fill threshold of $\sim 5.9\text{--}6.25\%$ ($= 1/16$), and the anisotropy-driven AIC branch is mapped flat at $\gamma/\omega_{ci} \approx 0.037 \pm 0.003$ across 0–10 % warm fill — a bounded, characterised plug-gate rather than a surprise. A 40,000-sample global UQ returns

$Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.000$ and zero failed evaluations, dominated by β_c — the quantity the stabilisation scheme protects. The one loss channel held outside that envelope, central-cell cross-field transport, is carried as an explicit, gyro-Bohm-bounded item. The core is MHD- and microstability-sound, with a quantified margin and an honest, separately-closed sideways-loss budget.

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Reproducibility. The stability, potential and robustness values reported here are frozen results (AG-DR-03, BU-L2-A5, BU-L2-A7, AG-CX-07, KX-L1-A5-BU, KX-L1-A18-BU, HX-23) in the Kronos Physics De-Risking Register, doi:10.5281/zenodo.22133057. This paper is archived at doi:10.5281/zenodo.22645854; official version: <https://www.kronosfusionenergy.com/publications/central-cell-radial-transport-and-interchange-ballooin.html> and its official version is hosted at <https://www.kronosfusionenergy.com/publications/central-cell-radial-transport-and-interchange-ballooin.html>. The figure-generation script (`make_figs_centralcell_258.py`), which reads only the frozen scalar results above, is archived with the deposit; all runs are seed-deterministic (seed 20260726).

Data availability

This paper is archived at doi:10.5281/zenodo.22645854. The PIC, kinetic-dispersion and Monte-Carlo UQ inputs and the figure scripts are archived with the deposit and reference the Kronos 2026 series and its Physics De-Risking Register (doi:10.5281/zenodo.22133057). Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The authors declare no other competing interests.

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